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Consumer Preferences and Willingness to Pay for Decarbonized Electricity Generation

A Choice-Based Conjoint Analysis of Japan and Germany

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Motivation and Research Question

- The transition to a **low-carbon electricity system** is a policy goal worldwide
- Meanwhile, in Japan, the retail electricity market was **fully liberalized** in April 2016: consumers can now freely choose their electricity company and plan

$\Rightarrow$ Decarbonization now also depends on **demand-side behavior**: which plans consumers choose

- A plan is a **bundle**: energy mix, price, supplier and other contract terms

Research question:

How do consumers evaluate **individual energy sources** and **contract attributes**, and how do these valuations compare between **Japan and Germany**?

Three gaps in the literature

1. Most evidence comes from a **single country**
2. Renewables are often treated as **one aggregate category**
3. The **relationship** between preferences for energy sources and for contract conditions has received **less attention**

Literature Review

We reviewed 30 stated-preference studies

- **Single-country studies: 23**
  Germany 10, Japan 8, US 2, Italy, Korea, Finland
  $\Rightarrow$ detailed contract attributes, but one country
- **Cross-country studies: 7**
  JP–US, JP–DE, KR–UK, US–UK, EU and OECD panels
  $\Rightarrow$ compare countries, but mix aggregated or tariffs only

Positioning in the Literature

▸ Full review of 30 studies: Appendix A7

Study	Country	Model	Int'l comp.	Energy mix	Elec. cost	Elec. provider	Env. label	Price guar.
Single-country studies (23)								
Kaenzig et al. (2013)	Germany	HB	×	△[a]	●	●	●	●
Tabi et al. (2014)	Germany	LC, HB	×	●	●	●	●	●
Danne et al. (2021)	Germany	G-MNL	×	●	●	×	●	●
Ouchi (2024)	Japan	CL	×	△[a]	●	●	●	●
19 more			⋮	⋮	⋮	⋮	⋮	⋮
Cross-country studies (7)								
Murakami et al. (2015)	Japan & US	RPL	●	●	●	×	×	×
Wu et al. (2022)	10 EU countries	MXL	●	●	●	●	×	×
Hassett et al. (2024)	9 OECD countries	CL	●	△[b]	●	●	×	×
Nakai et al. (2024)	Japan & Germany	MXL	●	×	×	×	×	△
Lim et al. (2025)	Korea & UK	MXL	●	×	●	●	×	×
2 more			⋮	⋮	⋮	⋮	⋮	⋮
This study	**Japan & Germany**	**MXL**	●	●	●	●	●	●

●: included; △: partly; ×: not included. [a] Effect of a 1% change cannot be estimated. [b] Calculated indirectly.

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Danne et al. (2021)	Germany	G-MNL	×	●	●	×	●	●
Ouchi (2024)	Japan	CL	×	△[a]	●	●	●	●
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Nakai et al. (2024)	Japan & Germany	MXL	●	×	×	×	×	△
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Danne et al. (2021)	Germany	G-MNL	×	●	●	×	●	●
Ouchi (2024)	Japan	CL	×	△[a]	●	●	●	●
19 more								
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Nakai et al. (2024)	Japan & Germany	MXL	●	×	×	×	×	△
Lim et al. (2025)	Korea & UK	MXL	●	×	●	●	×	×
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Positioning in the Literature

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Danne et al. (2021)	Germany	G-MNL	×	●	●	×	●	●
Ouchi (2024)	Japan	CL	×	△[a]	●	●	●	●
⋮ *19 more*			⋮	⋮	⋮	⋮	⋮	⋮
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Lim et al. (2025)	Korea & UK	MXL	●	×	●	●	×	×
⋮ *2 more*			⋮	⋮	⋮	⋮	⋮	⋮
This study	**Japan & Germany**	**MXL**	●	●	●	●	●	●

●: included; △: partly; ×: not included. [a] Effect of a 1% change cannot be estimated. [b] Calculated indirectly.

No previous study combines a **cross-country** design with **source-level** energy mix and **contract attributes** under a **common experiment**.

Why Compare Japan with Germany?

- **Policy leader**: ranked 14th in CCPI 2024, the highest among G7 countries (the Climate Change Performance Index 2024)
- **Early renewable support**:
  - EEG and feed-in tariff since 2000
  - feed-in premium since EEG 2014
  - achieved **High renewable share**: about 60% of generation, vs. about 30% globally
- **Comparable**: both are large economies with similar climate concern
  - "global emergency": 79% Japan, 77% Germany

Sources: Burck et al. (2023); Ember (2026); Energy Institute (2025); Flynn et al. (2021).

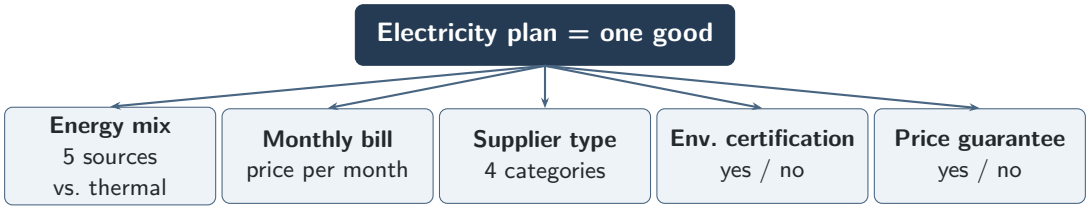
Method: Choice Experiment and Part-Worth Utilities

Contingent valuation (CVM)

Values a good **as a whole**
⇒ one WTP for the whole good
⇒ cannot tell **which features** drive the value

Choice experiment (CBC)

A good is a **bundle of attributes** (Lancaster, 1966)
⇒ respondents choose among bundles
⇒ each attribute's contribution to utility: **part-worth** β_k
⇒ WTP per attribute $= -\beta_k / \beta_{\text{price}}$



Deterministic utility (Eq. 3): $V_{ntj} = \mathbf{X}'_{ntj}\boldsymbol{\beta}_n + \beta_{\text{price}}\, p_{ntj}$

Choice Experiment Design: Attributes and Levels

▸ Orthogonal design: Appendix A8

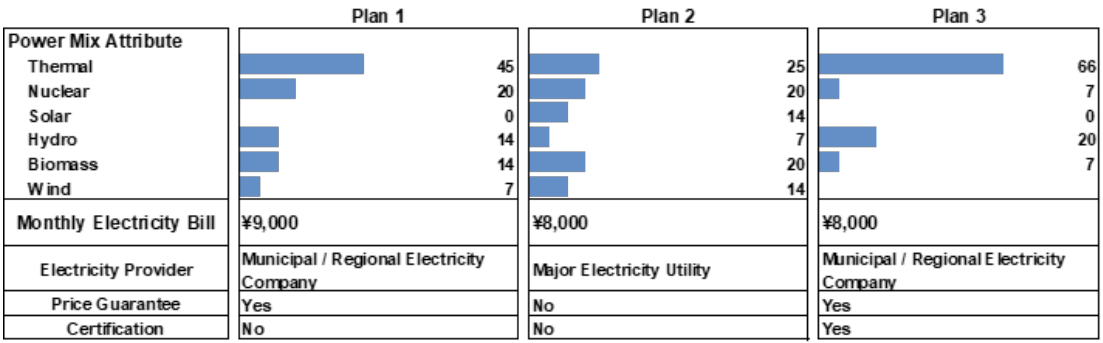
Attribute	Japan	Germany
Energy mix (share of each source; thermal = residual to 100%)		
Nuclear, solar, wind, hydro, biomass	0%, 7%, 14%, 20% each (same in both countries)	
Contract attributes		
Monthly bill	¥6,000 – ¥10,000 (5 levels, step ¥1,000)	€120 – €140 (5 levels, step €5)
Supplier	Major utility (ref.) Major new supplier Small/medium-sized new supplier Municipal/regional new supplier	Large supplier (ref.) Medium-sized supplier Municipal supplier Independent power marketer
Env. certification	No / Yes	No / Yes
Price guarantee	No / Yes	No / Yes

32 profiles (orthogonal main-effects design)

How to read the results (utility change per one unit): **energy mix: per 1 pp of thermal replaced by the source**; bill: per 1,000 JPY / 1 EUR; guarantee and label: yes vs. no; supplier: vs. the reference supplier.

Example of a Choice Task (Japan)

If you had to choose one of the following electricity plans, which plan would you be most likely to choose? Please select only one option.



Power Mix Attribute	Plan 1	Plan 2	Plan 3
Thermal	45	25	66
Nuclear	20	20	7
Solar	0	14	0
Hydro	14	7	20
Biomass	14	20	7
Wind	7	14	
Monthly Electricity Bill	¥9,000	¥8,000	¥8,000
Electricity Provider	Municipal / Regional Electricity Company	Major Electricity Utility	Municipal / Regional Electricity Company
Price Guarantee	Yes	No	Yes
Certification	No	No	Yes
	choice 1	choice 2	choice 3

Data Overview

Japan

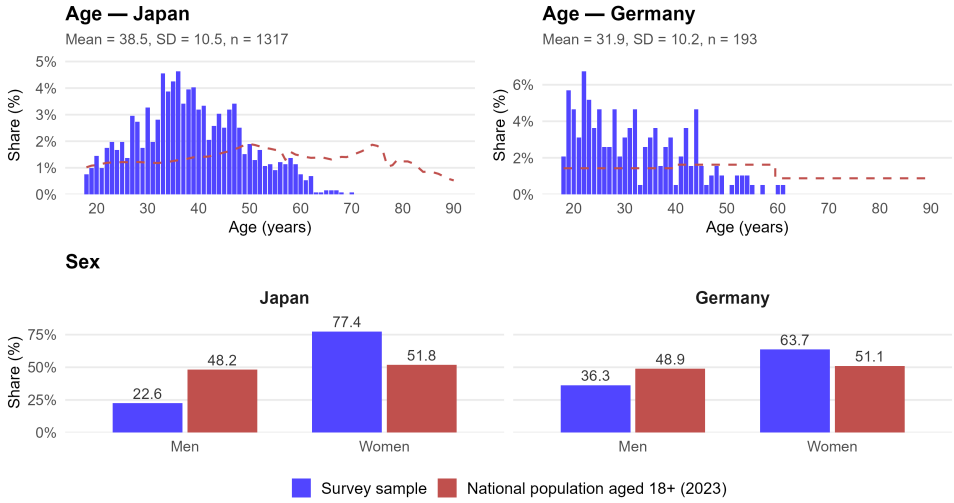
Fielded:	Dec 11, 2023
Vendor:	Yahoo! Crowdsourcing
Sample:	1,599 → **1,317**
Bill levels:	¥6,000–10,000
Reference:	avg. bill ¥8,390 (2023)
Median bill:	¥8,600 ⇒ **around** median

Germany

Fielded:	Jun 13–14, 2023
Vendor:	Clickworker
Sample:	367 → **193**
Bill levels:	€120–140
Reference:	kWh by household size × €0.413
Median bill:	€87 ⇒ **above** median

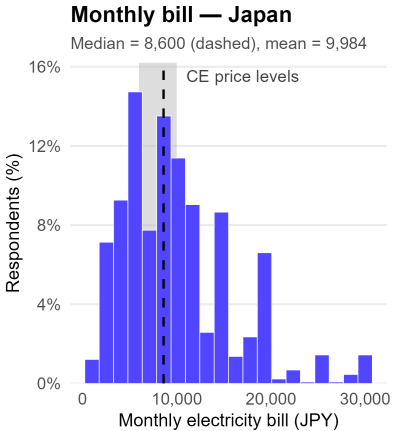
- Screening: consent, completion, known contract status, age ≥ 18, plausible bill
- Data source: Ouchi (2024), unpublished.

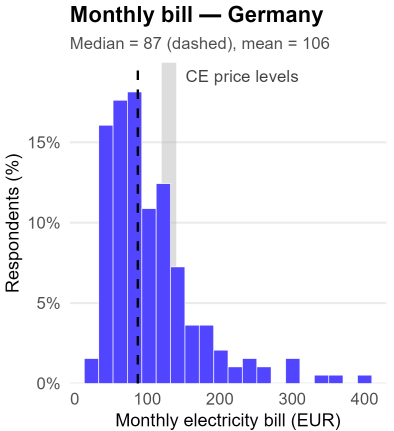
Sample Characteristics: Age and Sex

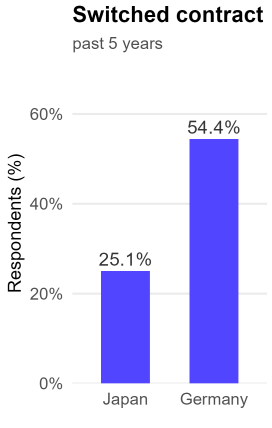


Bars: survey sample. Red: national population aged 18+ (dashed line in age panels; Germany spread evenly within 18–39 / 40–59 / 60+). Japan: Statistics Bureau, 1 Oct 2023. (

Sample Characteristics: Electricity Bill and Switching







Econometric Model: From Utility to Likelihood

1 **Random utility**
choose the highest-utility plan

$$U_{ntj} = V_{ntj} + \varepsilon_{ntj}$$
n chooses j if $U_{ntj} > U_{nti}\ \forall i \neq j$

2 **Deterministic utility**
non-price attributes + price (Eq. 3)

$$V_{ntj} = \mathbf{X}'_{ntj}\boldsymbol{\beta}_n + \beta_{\text{price}}\, p_{ntj}$$

3 **Choice probability (logit)**
ε i.i.d. Gumbel; 3 plans

$$P_{ntj} = \frac{\exp(V_{ntj})}{\sum_{i=1}^{3} \exp(V_{nti})}$$

4 **Mixed logit**
tastes vary: $\boldsymbol{\beta}_n \sim N(\boldsymbol{\beta}, \boldsymbol{\Omega})$

$$P_n = \int \prod_t \prod_j P_{ntj}(\boldsymbol{\beta}_n)^{y_{ntj}}\, f(\boldsymbol{\beta}_n)\, d\boldsymbol{\beta}_n$$

$y_{ntj} = 1$ if plan j is chosen; 11 tasks per respondent (last task: 2 plans).
Price coefficient fixed; other coefficients random, $\boldsymbol{\Omega}$ full (correlated) or diagonal.

Model Selection: Likelihood-Ratio Tests and Fit

▸ Full fit statistics: Appendix A3

		Japan			Germany		
Test	**Restriction**	**LR**	**d.f.**	***p***	**LR**	**d.f.**	***p***
(1) vs (2)	No random coefficients (CL)	986.5	10	<0.001	195.6	10	<0.001
(2) vs (3)	Fewer random coefficients (uncorr.)	≤ 0	5	–	1.5	4	0.820
(2) vs (4)	No correlation (all random)	789.7	45	<0.001	178.5	45	<0.001
(3) vs (5)	No correlation (selected random)	567.3	10	<0.001	97.4	15	<0.001
(5) vs (4)	Fewer random coefficients (corr.)	219.4	40	<0.001	82.6	34	<0.001

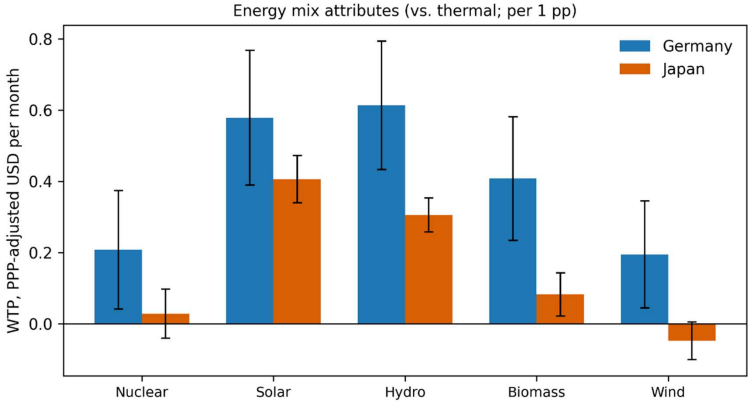
			AIC		BIC	
Model	**Random coef.**	**Corr.**	**JP**	**DE**	**JP**	**DE**
(1) CL	none	–	24,294	3,444	24,378	3,506
(2) MXL	all	no	23,328	3,268	23,487	3,387
(3) MXL	selected†	no	23,315	3,262	23,436	**3,358**
(4) MXL	**all**	**yes**	**22,628**	**3,180**	23,128	3,553
(5) MXL	selected†	yes	22,767	3,194	**22,964**	3,375

Main specification: Model (4)

- LR tests **reject every restriction** of (4): correlation and full random set
- **Lowest AIC** in both countries
- BIC favors the sparser (5)/(3): (4) has 66 parameters

† SD significant in (2). Red: best value. LR $= 2(\mathrm{LL}_u - \mathrm{LL}_r)$; SD $= 0$ restrictions lie on the boundary, so p-values are conservative. (2) vs (3), Japan: simulation error.

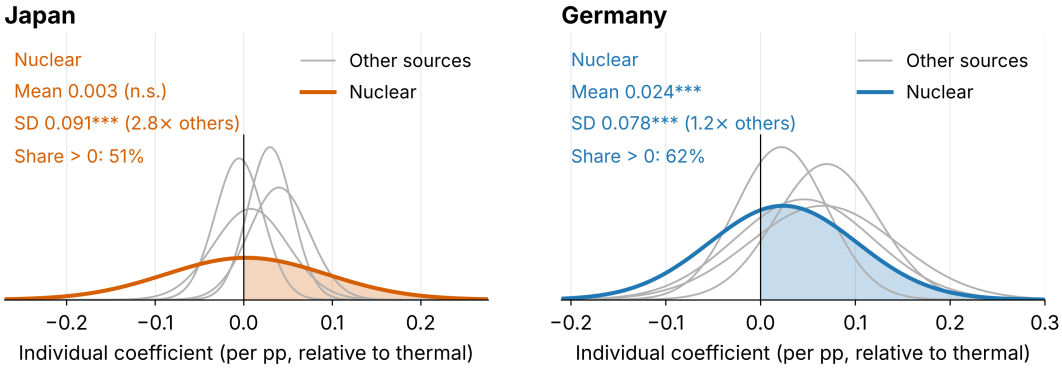
Result 1: WTP for Energy Sources (relative to thermal)



Model (4), PPP-adjusted USD per month per pp (121 JPY/USD; 0.860 EUR/USD); 95% CI.

Solar and hydro valued most in both; **nuclear** positive only in **DE**; **wind** lowest (negative in **JP**).

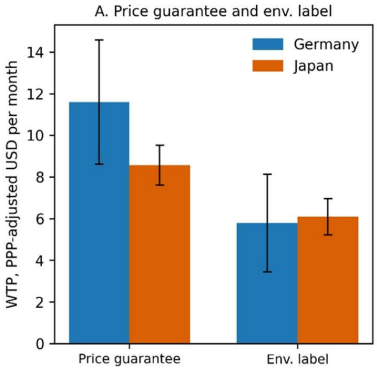
Result 2: Preference Heterogeneity — Nuclear Divides Japan



Implied distribution of individual coefficients, $N(\beta_k, \sigma_k^2)$, Model (4). Shaded: share with a positive coefficient. Horizontal axes differ by country.

Result 3: Contract Attributes Matter alongside the Mix

▸ Correlations: A4–A5

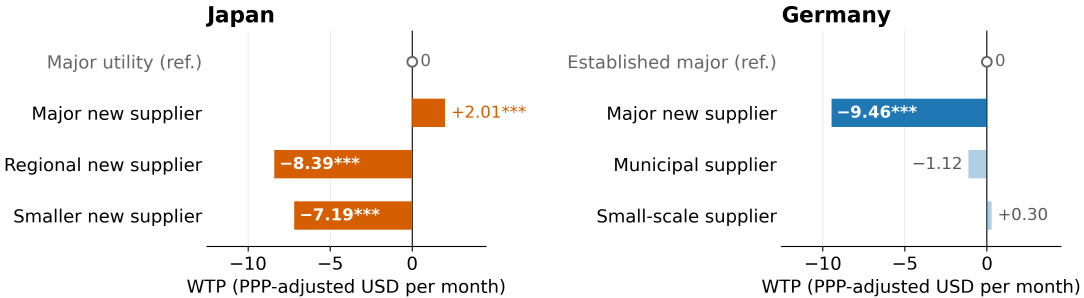


Model (4), PPP-adjusted USD per month; 95% CI.

- **Price guarantee**: the most valued contract attribute in both countries
- **Label / guarantee ratio**: **DE** 0.50, **JP** 0.71
  ⇒ **JP** places somewhat more weight on the label (10% level)
- Respondents who value the guarantee more tend to value **biomass and wind** (vs. thermal) **less**, in both countries

Contract terms are valued **alongside the energy mix**.

Result 4: Preference for Established Suppliers

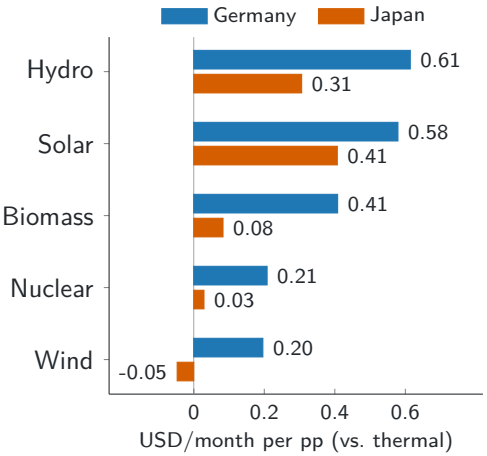


WTP relative to the reference supplier, whose value is normalized to 0 (dummy coding), PPP-adjusted USD per month, Model (4). Dark: $p < 0.01$; pale: not significant.

In both countries, some new entrants are valued **below the established / major supplier** — consistent with **trust in suppliers** (not tested directly).

Summary

Main result: PPP-adjusted WTP



Model (4). **JP** nuclear: not significant.

1. **Solar and hydro** are valued most and **wind** least in both countries
2. **Nuclear**: positive in **DE**; in **JP** near zero and **divided**
   SD about 2.8× that of other sources; about half value it positively
3. **Price guarantee** is the most valued contract attribute (**DE** 11.59, **JP** 8.56 USD/month)
   JP weights the label somewhat more (10% level)
4. **Established suppliers** are valued in both countries

Answer to the Research Question

RQ: How do consumers evaluate energy sources and contract attributes, and how do they compare between Japan and Germany?

	Germany		Japan
Energy sources (vs. thermal)			
Order of non-thermal sources	Hydro > solar > biomass > nuclear > wind	≈	Solar > hydro > biomass > nuclear > wind
Nuclear	**Positive** (0.21)	≠	Near zero; **divided** (SD 2.8× other sources)
Contract terms			
Most valued	Price guarantee (11.59)	≈	Price guarantee (8.56)
Label / guarantee	0.50	≠	0.71 (difference at 10% level)
Guarantee vs. biomass, wind	Negative correlation	≈	Negative correlation
Suppliers	Established valued	≈	Established valued

≈ similar ≠ different. WTP in PPP-adjusted USD per month (energy sources: per pp), Model (4). Correlations: Appendix A4–A5.

Limitations

- Samples are younger and more female; German N = 193
- Price levels sit at different positions relative to actual bills
- Forced-choice stated preferences; mechanisms not identified
- **Next**: representative samples with comparable price settings; interactions of contract terms and energy sources

Thank you for listening

Any comments and questions are welcome

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Reply to the Discussant (1/4): Respondent Characteristics

We thank the discussant for the careful and constructive comments.

Q1 (discussant)	A (our reply)
Income: an income effect on WTP? Data on income?	Collected; descriptive statistics in Appendix A10. Income $\times$ price / attribute interactions not yet estimated $\rightarrow$ future work
Solar PV: bills may not match demand if households own solar panels	PV ownership was not collected; prosumers may evaluate contracts differently $\rightarrow$ stated as a limitation
Location: climate may drive electricity demand	Respondents' region was not collected $\rightarrow$ stated as a limitation
Hypothetical bill: can respondents imagine an "average household" payment?	The average bill is a common, objective reference point; the price range covers actual variation in expenditure. Possible hypothetical bias $\rightarrow$ limitation
Representativeness: include more individual characteristics	Samples not representative; estimates not generalized (stated in the paper). Random parameters capture part of the heterogeneity; observed characteristics as interactions $\rightarrow$ future work

Reply to the Discussant (2/4): Conceptual Model

Q2. Respondents may distinguish sources by supply stability, while the price guarantee is itself an attribute. Make explicit that source WTPs exclude price-stability differences; the same applies to suppliers. How do the guarantee and certification affect utility (e.g., expected utility)? A conceptual model is needed.

A. Agreed.

- "Stability" enters in three places: the **price guarantee** (contractual), **energy sources** (perceived supply / price stability), and **suppliers** (perceived reliability)
- The guarantee is varied independently of the mix and the supplier → source and supplier WTPs are **net of the contractual guarantee**, but may still contain **perceived stability**; our main-effects design cannot separate the two
- We will state this explicitly and add a **conceptual model** (e.g., guarantee and certification raise expected utility by reducing price risk and quality uncertainty)
- Separating them requires guarantee × source / supplier **interactions** → future work

Reply to the Discussant (3/4): Germany and Nuclear Power

Q3. The German market description relies mainly on Kaenzig et al. (2013); much must have changed between 2013 and 2023.

A. Agreed.

- The 2022 energy price shock and the 2023 nuclear phase-out are noted in Sec. 5.1; we will bring them into the market description
- This context may also help interpret the high valuation of the price guarantee in Germany

Q4. "Nuclear power divides Japanese consumers" — why not a latent class analysis?

A.

- The MXL already estimates the full distribution of nuclear preferences, with substantial shares on both sides (Result 2)
- A latent class model is a useful complementary check of distinct segments $\rightarrow$ future work

Reply to the Discussant (4/4): Policy Implications

Q5. The discussion of policy implications is thin. For example, compute national aggregates of WTP, or compare WTP with actual differences in generation costs.

A. Thank you; we agree we can go further.

- **Current paper**:
  - consider demand-side acceptance in decarbonization policy
  - offer price-stability options alongside low-carbon plans
  - offer a menu of contracts rather than a single design
- **Extension**: compare WTP relative to thermal with differences in generation costs
- **National aggregates**: require weighting, as the samples are not representative

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[6] Ahmed, B. M., & Fikru, M. G. (2025). Evaluating consumer preference and willingness to pay for low emission electricity: Evidence from a national choice experiment in the US. Energy Economics, 148, Article 108650. https://doi.org/10.1016/j.eneco.2025.108650

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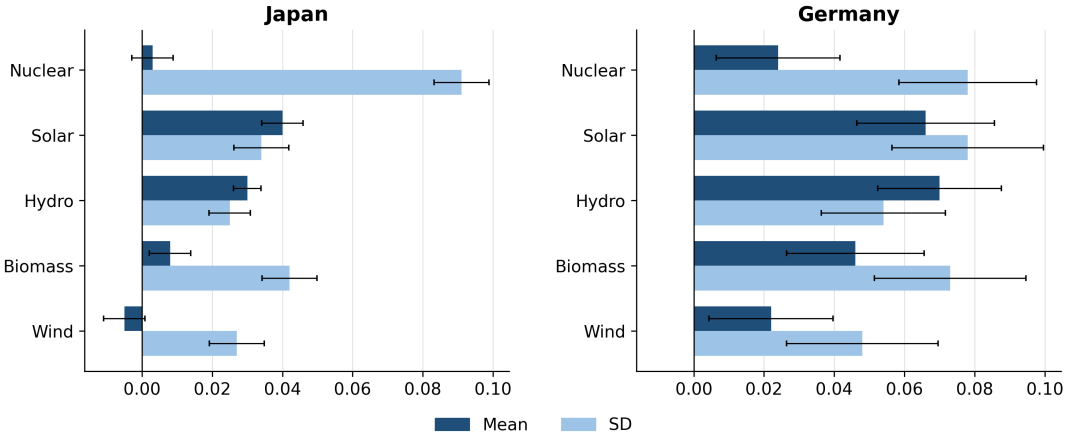
Appendix

A1. Estimation Results: Model (4)

Japan					Germany				
	Mean	s.e.	SD	s.e.		Mean	s.e.	SD	s.e.
Nuclear	0.003	0.003	0.091***	0.004	Nuclear	0.024***	0.009	0.078***	0.010
Solar	0.040***	0.003	0.034***	0.004	Solar	0.066***	0.010	0.078***	0.011
Hydro	0.030***	0.002	0.025***	0.003	Hydro	0.070***	0.009	0.054***	0.009
Biomass	0.008***	0.003	0.042***	0.004	Biomass	0.046***	0.010	0.073***	0.011
Wind	−0.005*	0.003	0.027***	0.004	Wind	0.022**	0.009	0.048***	0.011
Price	−0.810***	0.018			Price	−0.132***	0.011		
Guarantee	0.839***	0.047	0.833***	0.050	Guarantee	1.314***	0.166	1.425***	0.154
Label	0.597***	0.044	0.269***	0.063	Label	0.657***	0.133	0.477***	0.167
Major new	0.197***	0.055	0.542***	0.083	Major new	−1.072***	0.200	1.454***	0.232
Regional new	−0.823***	0.058	1.248***	0.064	Municipal	−0.127	0.189	0.803***	0.190
Smaller new	−0.704***	0.059	1.319***	0.063	Small-scale	0.034	0.136	0.685***	0.188
Log-lik.	−11,247.97	(14,487 choices)			Log-lik.	−1,523.77	(2,123 choices)		

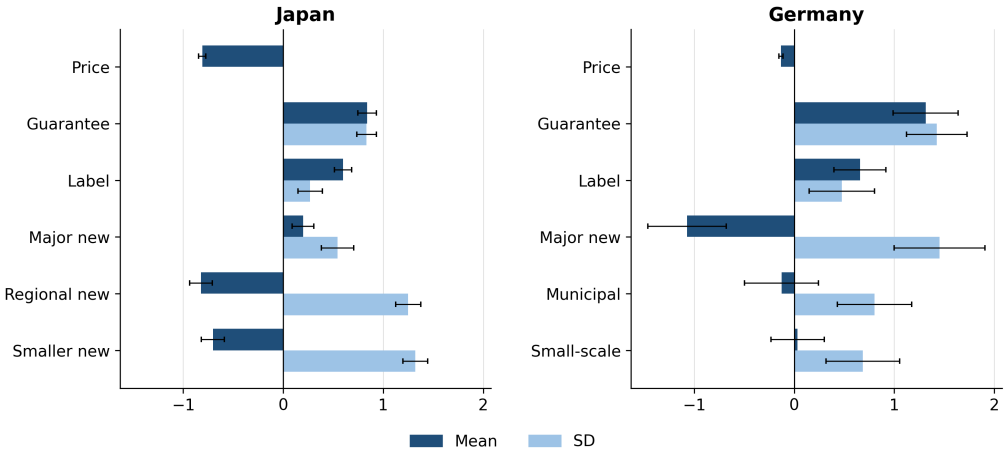
Price per 1,000 JPY (Japan) and per 1 EUR (Germany). Energy sources relative to thermal. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A1. Estimation Results: Model (4) — Energy Sources



Dark: mean; light: SD of random coefficients. Whiskers: 95% CIs. Relative to thermal.

A1. Estimation Results: Model (4) — Other Attributes



Dark: mean; light: SD of random coefficients. Whiskers: 95% CIs. Price per 1,000 JPY (Japan) and per 1 EUR (Germany).

A2. Willingness to Pay: Local Currency and PPP-adjusted USD

	Local currency		PPP-adjusted USD	
	Germany (EUR)	Japan (JPY)	Germany	Japan
Energy mix (per pp, relative to thermal)				
Nuclear	0.179**	3.46	0.208**	0.029
Solar	0.498***	49.20***	0.579***	0.407***
Hydro	0.528***	37.03***	0.614***	0.306***
Biomass	0.351***	10.03***	0.408***	0.083***
Wind	0.168**	−5.74*	0.196**	−0.047*
Contract attributes (monthly)				
Price guarantee	9.97***	1,035.9***	11.59***	8.56***
Env. certification label	4.98***	736.9***	5.79***	6.09***
DE: Major new supplier	−8.13***		−9.46***	
DE: Municipal supplier	−0.97		−1.12	
DE: Small-scale supplier	0.26		0.30	
JP: Major new supplier		243.4***		2.01***
JP: Regional supplier		−1,015.5***		−8.39***
JP: Smaller supplier		−869.5***		−7.19***

PPP: OECD household final consumption expenditure PPPs, 2023, OECD base (0.860 EUR/USD, Germany; 121 JPY/USD, Japan).

A3. Model Fit and Likelihood Ratio Tests

Back

	Japan					Germany				
Model	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
Log-lik.	−12136.1	−11642.8	−11641.3	−11248.0	−11357.7	−1710.8	−1613.0	−1613.8	−1523.8	−1565.1
Params	11	21	16	66	26	11	21	17	66	32
AIC	24294.2	23327.6	23314.6	**22627.9**	22767.3	3443.7	3268.1	3261.6	**3179.5**	3194.2
BIC	24377.6	23486.8	23435.9	23128.3	**22964.4**	3505.9	3386.9	**3357.8**	3553.1	3375.3
Hit rate (%)	62.57	62.57	62.57	62.57	62.57	64.86	64.86	64.86	65.47	64.86

		Japan			Germany		
Models	Restriction tested	LR	d.f.	p	LR	d.f.	p
(1) vs (2)	CL vs MXL (uncorrelated, all random)	986.54	10	< 0.001	195.62	10	< 0.001
(2) vs (3)	All vs selected random (uncorrelated)	≤ 0	5	–	1.54	4	0.820
(2) vs (4)	Uncorrelated vs correlated (all random)	789.70	45	< 0.001	178.52	45	< 0.001
(3) vs (5)	Uncorrelated vs correlated (selected)	567.28	10	< 0.001	97.42	15	< 0.001
(5) vs (4)	Selected vs all random (correlated)	219.40	40	< 0.001	82.64	34	< 0.001

Boundary restrictions (SD = 0): χ^2-based p-values are conservative. In Japan, (3) attains a slightly higher simulated log-likelihood than (2) (simulation error).

A4. Correlations among Random Parameters: Japan, Model (4)

▸ Reading: A4b

	Nuclear	Solar	Hydro	Biomass	Wind	Guar.	Label	Major new	Regional	Smaller
Nuclear	–	0.251**	0.038	−0.316***	−0.857***	0.112*	0.269	−0.059	−0.558***	−0.411***
		(0.115)	(0.107)	(0.078)	(0.094)	(0.066)	(0.191)	(0.134)	(0.045)	(0.049)
Solar		–	0.374***	0.519***	−0.237	−0.115	0.103	−0.450***	−0.461***	−0.098
			(0.142)	(0.125)	(0.181)	(0.114)	(0.249)	(0.160)	(0.097)	(0.113)
Hydro			–	0.465***	−0.287*	0.275**	0.393*	0.481***	0.081	0.482***
				(0.122)	(0.173)	(0.108)	(0.210)	(0.166)	(0.112)	(0.096)
Biomass				–	0.470***	−0.608***	−0.013	0.193	−0.100	0.310***
					(0.142)	(0.097)	(0.209)	(0.151)	(0.080)	(0.075)
Wind					–	−0.546***	−0.410*	0.048	0.382***	0.274**
						(0.096)	(0.241)	(0.196)	(0.109)	(0.109)
Guar.						–	0.441**	−0.025	0.141*	−0.045
							(0.173)	(0.140)	(0.080)	(0.076)
Label							–	0.375	−0.589***	−0.511***
								(0.255)	(0.169)	(0.185)
Major new								–	0.169	0.300**
									(0.127)	(0.123)
Regional									–	0.849***
										(0.028)

Computed from the Cholesky factor; delta-method s.e. in parentheses. Correlations among energy sources partly reflect heterogeneity in the valuation of thermal power. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A4b. What the Correlations Suggest: Japan

Back to A4

- **Nuclear vs. wind**: strong trade-off (-0.86^{***}); nuclear is also negatively correlated with biomass (-0.32^{***})
  $\Rightarrow$ those who value nuclear more value wind (and biomass) less
- **Renewables move together**: solar–biomass 0.52^{***}, hydro–biomass 0.47^{***}, biomass–wind 0.47^{***}, solar–hydro 0.37^{***}
- **Nuclear and the label vs. new suppliers**: nuclear–regional -0.56^{***}, nuclear–smaller -0.41^{***}; label–regional -0.59^{***}, label–smaller -0.51^{***}
  $\Rightarrow$ those who value nuclear or the label more discount regional and smaller new suppliers more
- **Guarantee and label go together** (0.44^{**})
- **Regional and smaller new suppliers** are valued alike (0.85^{***})
  $\Rightarrow$ a common attitude toward small or regional new entrants

Caution: exploratory; not discussed in the paper. Energy sources share the thermal reference and suppliers share the major-utility reference, so positive correlations within each group partly reflect heterogeneity in valuing the reference.

A5. Correlations among Random Parameters: Germany, Model (4)

Reading: A5b

	Nuclear	Solar	Hydro	Biomass	Wind	Guar.	Label	Major new	Municipal	Small-scale
Nuclear	–	0.180	0.680***	−0.115	0.084	0.056	−0.166	−0.482***	0.254	−0.114
		(0.161)	(0.138)	(0.164)	(0.195)	(0.136)	(0.312)	(0.114)	(0.273)	(0.248)
Solar		–	0.641***	0.613***	0.450**	−0.696***	−0.214	−0.628***	−0.656***	−0.752***
			(0.153)	(0.142)	(0.185)	(0.096)	(0.326)	(0.134)	(0.206)	(0.192)
Hydro			–	0.518***	0.123	−0.121	0.001	−0.818***	−0.205	−0.632**
				(0.176)	(0.232)	(0.217)	(0.322)	(0.089)	(0.303)	(0.247)
Biomass				–	0.284	−0.482***	0.358	−0.643***	−0.466**	−0.471**
					(0.229)	(0.148)	(0.284)	(0.126)	(0.232)	(0.231)
Wind					–	−0.645***	−0.636**	−0.501**	−0.590**	−0.414
						(0.157)	(0.275)	(0.211)	(0.236)	(0.311)
Guar.						–	0.085	0.253	0.402*	0.302
							(0.303)	(0.159)	(0.221)	(0.260)
Label							–	0.206	0.499*	0.388
								(0.308)	(0.295)	(0.370)
Major new								–	0.567**	0.629***
									(0.229)	(0.206)
Municipal									–	0.597**
										(0.269)

Computed from the Cholesky factor; delta-method s.e. in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A5b. What the Correlations Suggest: Germany

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- **Renewables move together**: solar–hydro 0.64^{***}, solar–biomass 0.61^{***}, hydro–biomass 0.52^{***}, solar–wind 0.45^{**}
- **Nuclear goes with hydro** (0.68^{***}), not against the renewables (unlike nuclear–wind in Japan)
- **Renewables vs. non-reference suppliers**: hydro–major new -0.82^{***}, solar–small-scale -0.75^{***}, solar–municipal -0.66^{***}, biomass–major new -0.64^{***}
  $\Rightarrow$ those who value renewables more discount suppliers other than the large supplier more
- **Guarantee vs. solar** (-0.70^{***}), in addition to biomass and wind (in the paper)
- **Supplier types move together**: major new–small-scale 0.63^{***}, municipal–small-scale 0.60^{**}, major new–municipal 0.57^{**}

Caution: exploratory; not discussed in the paper; German N $= 193$. Energy sources share the thermal reference and suppliers share the large-supplier reference, so positive correlations within each group partly reflect heterogeneity in valuing the reference.

A6. Setting the Price Attribute

Two common approaches

- **Pivot design**: ±% around each respondent's reported bill — realistic, but sensitive to reporting errors
- **Fixed levels**: common levels around a reference household — simple, but ignores variation across households

Our approach: fixed levels centred on an objective national reference, with a range covering the realistic spread of household bills.

Japan: average household bill ¥8,390 (Jul–Sep 2023, FIES) ⇒ ¥6,000–10,000 in steps of 1,000

Germany: annual kWh by household size (co2online) × €0.413/kWh (Eurostat, H1 2023) ÷ 12; 3–4 person households hold ∼21% of households but ∼30% of consumption ⇒ €120–140 in steps of 5

PPP conversion is used only as a supplementary indicator, not to equalize levels.

A7. Literature Review (1/3): Single-Country Studies (23)

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Year	Country	Author	CE	Model	Energy mix	Elec. cost	Elec. provider	Env. label	Price guar.
1997	US (8 states)	Roe et al. (2001)	•	OLS	•	•	×	×	×
2000	Japan	Nomura & Akai (2004)	×	Weibull MLE	×	•	×	×	×
2005	Japan (Tokyo)	Ise (2006)	×	–	×	×	×	×	×
2008	Germany	Grösche & Schröder (2011)	•	RPL	•	•	×	×	×
2009	Germany	Kaenzig et al. (2013)	•	HB	△[a]	•	•	•	•
2009	Italy	Cicia et al. (2012)	•	LC	•	•	×	×	×
n.d.	Germany	Liebe et al. (2012)	•	Logit[e]	×	•	×	×	×
2012	Germany	Sagebiel et al. (2014)	•	CL, LC	△[d]	•	•	×	×
2012	Japan	Morita & Managi (2015)	•	MNL, LC	•	•	△[c]	×	×
n.d.	Germany	Dütschke & Paetz (2013)	•	Conjoint	×	•	×	×	△
2013	Germany	Tabi et al. (2014)	•	LC, HB	•	•	•	•	•
2014	Japan	Nakatani et al. (2015)	•	CL	•	•	×	×	×
2023	**Japan & Germany**	**This study**	•	**MXL**	•	•	•	•	•

Year = survey year (n.d.: not reported). •: included; △: partly; ×: not included. [a] Effect of a 1% change cannot be estimated. [c] Stability included as an alternative. [d] Aggregate renewable share only. [e] Test–retest design; landscape effects of onshore wind power.

A7. Literature Review (2/3): Single-Country Studies (23, cont.)

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Year	Country	Author	CE	Model	Energy mix	Elec. cost	Elec. provider	Env. label	Price guar.
2014	Germany	Rommel et al. (2016)	•	MXL	•	•	•	×	×
2014	Korea	Huh et al. (2015)	•	MXL	△[a]	•	•	×	×
n.d.	Germany	Herbes et al. (2015)	×	Neuropricing	×	•	×	×	×
2015–16	Finland	Ruokamo et al. (2019)	•	MNL, MXL	×	•	×	×	△
2016	Japan	Ida & Murakami (2016)	•	RPL	•	•	•	×	△
2016	Germany	Danne et al. (2021)	•	G-MNL	•	•	×	•	•
n.d.	Germany	Schlereth et al. (2018)	•	HB	×	×	×	×	•
n.d.	Japan	Nakano et al. (2018)	×	SEM, OP	×	•	×	×	×
2020	Japan	Yoshida (2021)	•	BWS, RPL	•	×	×	×	×
2023	Japan	Ouchi (2024)	•	CL	△[a]	•	•	•	•
2024	US	Ahmed & Fikru (2025)	•	CL, MXL	•	•	•	×	×
2023	**Japan & Germany**	**This study**	•	**MXL**	•	•	•	•	•

Year = survey year (n.d.: not reported). •: included; △: partly; ×: not included. [a] Effect of a 1% change cannot be estimated. BWS: best–worst scaling; SEM: structural equation model; OP: ordered probit.

A7. Literature Review (3/3): Cross-Country Studies (7)

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Year	Countries	Author	CE	Model	Energy mix	Elec. cost	Elec. provider	Env. label	Price guar.
2012–13	Japan & US	Murakami et al. (2015)	•	RPL	•	•	×	×	×
2014	US & UK	Buryk et al. (2015)	•	CL, MXL	×	•	×	×	△
2020	10 EU countries	Wu et al. (2022)	•	MXL	•	•	•	×	×
2020–21	Japan & Germany	Nakai et al. (2024)	•	MXL	×	×	×	×	△
2022	9 OECD countries	Hassett et al. (2024)	•	CL	△[b]	•	•	×	×
2022	4 EU countries	Mey et al. (2024)	•	MNLHB	•	•	×	×	×
2023	Korea & UK	Lim et al. (2025)	•	MXL	×	•	•	×	×
2023	**Japan & Germany**	**This study**	•	**MXL**	•	•	•	•	•

Year = survey year. •: included; △: partly; ×: not included. [b] Calculated indirectly.

Cross-country studies either aggregate the energy mix or focus on tariffs; none covers source-level energy mix together with environmental labels and price guarantees.

A8. Experimental Design: Orthogonal Main-Effects Plan

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Why a fractional design?

- Full factorial: 4^5 (shares) $\times 5$ (bill) $\times 4$ (supplier) $\times 2 \times 2 =$ **81,920** profiles
- An **orthogonal array** selects a fraction in which every pair of attribute levels appears in fixed proportions
- $\Rightarrow$ attributes are **uncorrelated** across profiles, so each **main effect** (part-worth) is identified independently

Our design

- ORTHOPLAN (IBM SPSS): **32 profiles**
- $10 \times 3 + 1 \times 2 = 32$: each respondent sees every profile once
- Same design in both countries; bill levels and supplier labels are country-specific

Energy mix and the thermal reference

Five shares at 0, 7, 14, 20% $\Rightarrow$ sum 0–100%. Thermal $= 100\% -$ sum $\in [0, 100]\%$ is not a separate attribute,
$\Rightarrow$ each source's part-worth is measured **relative to thermal**.

Limitations

- Main effects only: **interactions** (e.g., mix $\times$ guarantee) are not identified
- Orthogonal profiles do not guarantee statistically **efficient choice sets**

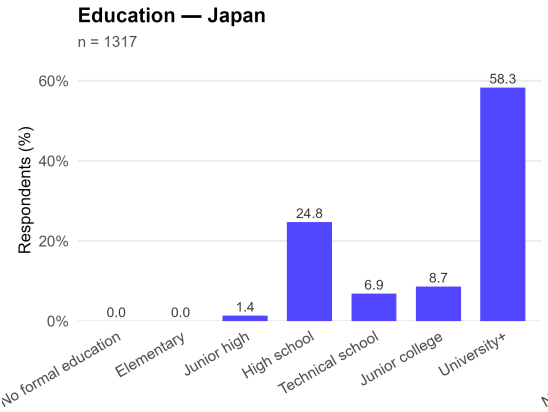
Addelman (1962); Abe (1993); Hedayat et al. (1999); Taguchi (2010); Tanaka & Horino (2011); Lin & Stufken (2025).

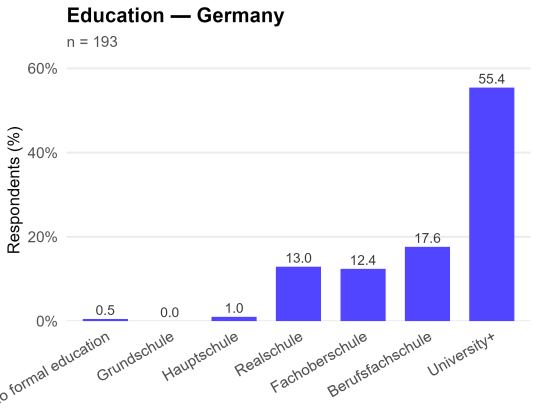
A9. Sample Characteristics: Summary Table

	Japan		Germany	
	Sample	Population	Sample	Population
N	1,317	–	193	–
Age, mean (SD)	38.5 (10.5)	–	31.9 (10.2)	–
Age 18–39	57.1%	26.5%	76.2%	31.5%
Age 40–59	40.2%	32.6%	22.8%	32.5%
Age 60+	2.7%	40.9%	1.0%	36.1%
Women	77.4%	51.8%	63.7%	51.1%
Monthly bill, mean (SD)	¥9,984 (5,970)	–	€106.2 (65.0)	–
Monthly bill, median	¥8,600	–	€87	–
Switched contract (past 5 yrs)	25.1%	–	54.4%	–

Population: residents aged 18+, Statistics Bureau of Japan (1 Oct 2023); Destatis (31 Dec 2023, 2022 census basis). Goodness-of-fit vs. population: all $p < 0.001$.

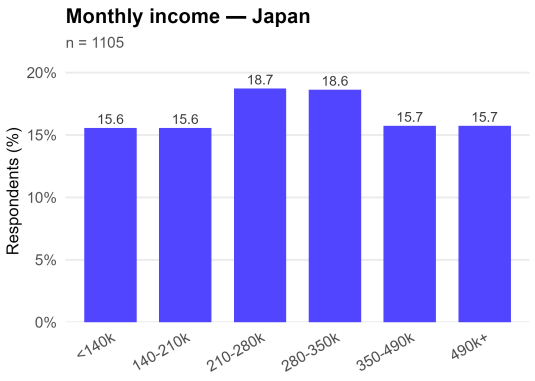
A10. Sample Characteristics (1/5): Education

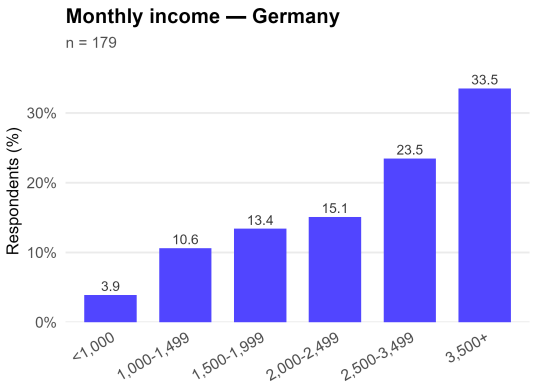




Share of respondents by highest completed education. Categories follow each country's school system.

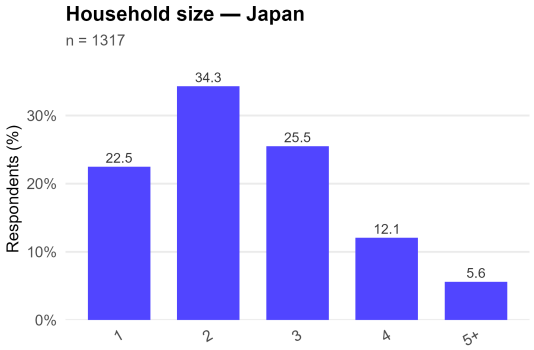
A10. Sample Characteristics (2/5): Monthly Income

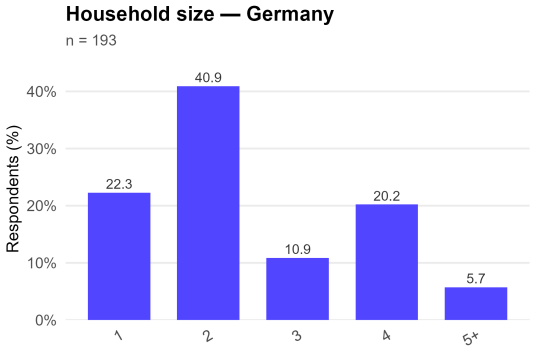




Japan: JPY; Germany: EUR. Respondents who did not report income are excluded (Japan $n = 1{,}105$; Germany $n = 179$).

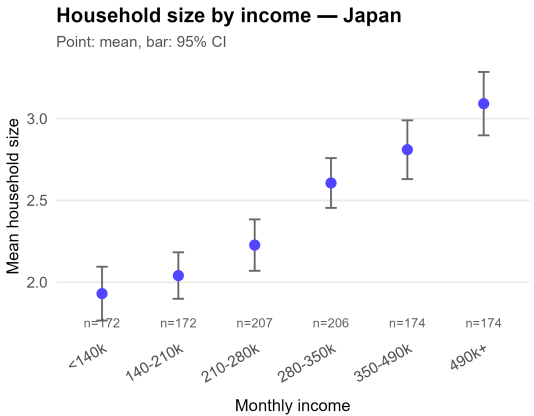
A10. Sample Characteristics (3/5): Household Size

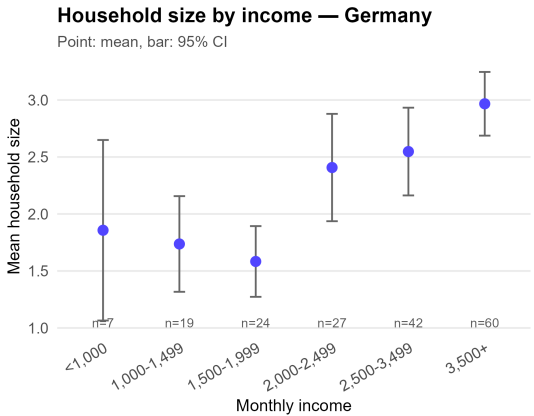




Number of household members, including the respondent.

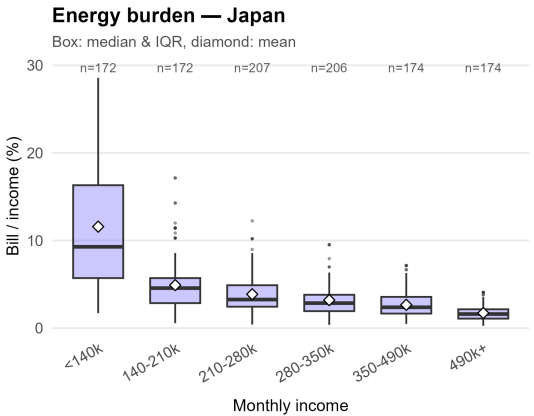
A10. Sample Characteristics (4/5): Household Size by Income

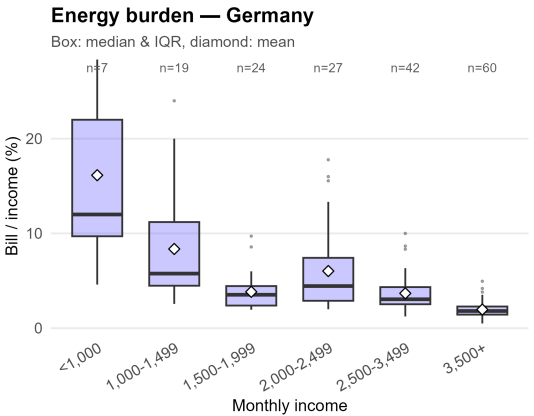




Point: mean household size; bar: 95% CI. Germany: small cells in the lowest income groups.

A10. Sample Characteristics (5/5): Energy Burden by Income





Energy burden = monthly electricity bill / monthly income (%). Box: median and IQR; diamond: mean.